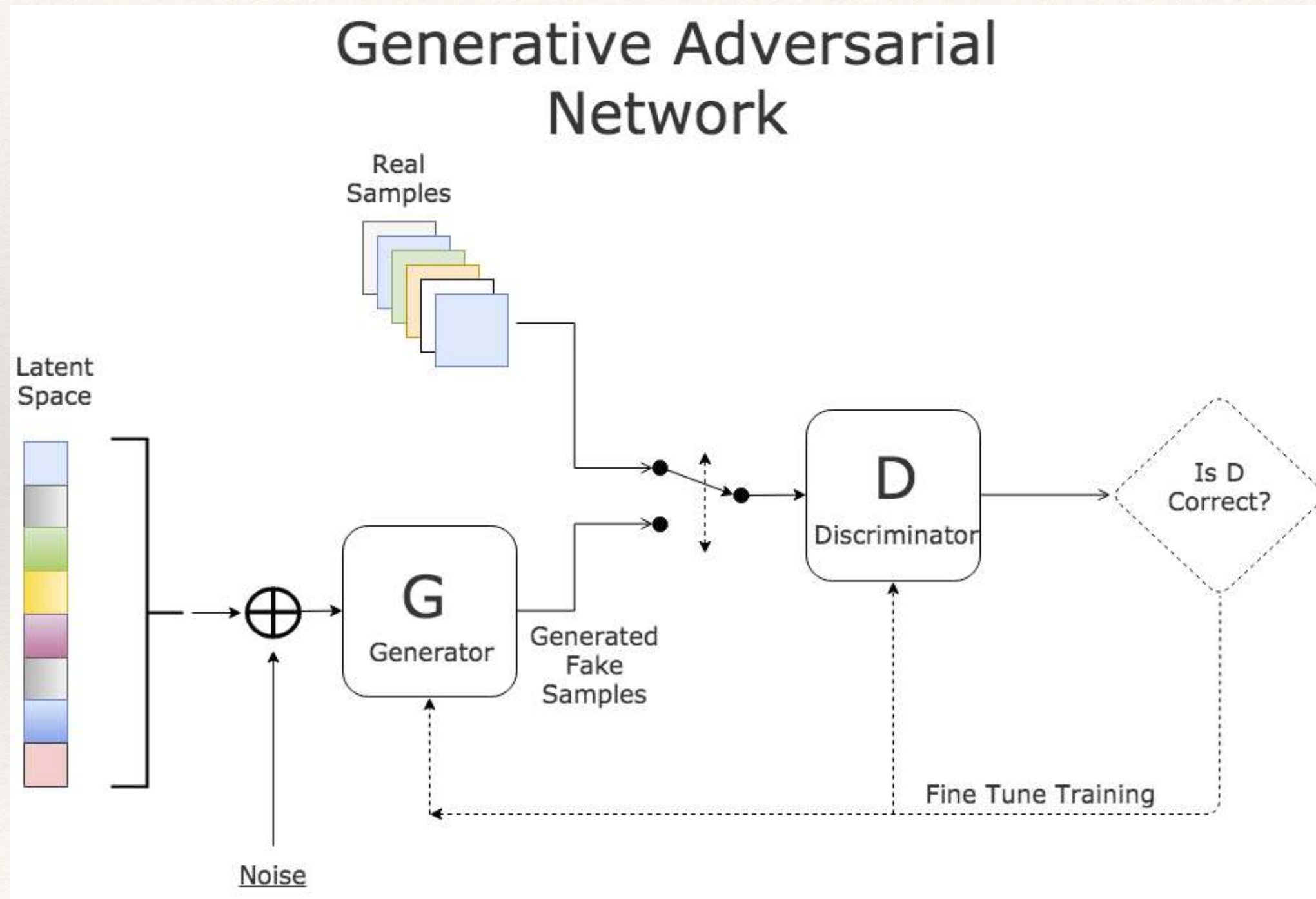


E9 205 Machine Learning for Signal Processing

Unsupervised Learning & Deep
Learning for Text

18-11-2019

GANs



GAN

Generative Adversarial Nets

**Ian J. Goodfellow*, Jean Pouget-Abadie†, Mehdi Mirza, Bing Xu, David Warde-Farley,
Sherjil Ozair‡, Aaron Courville, Yoshua Bengio§**

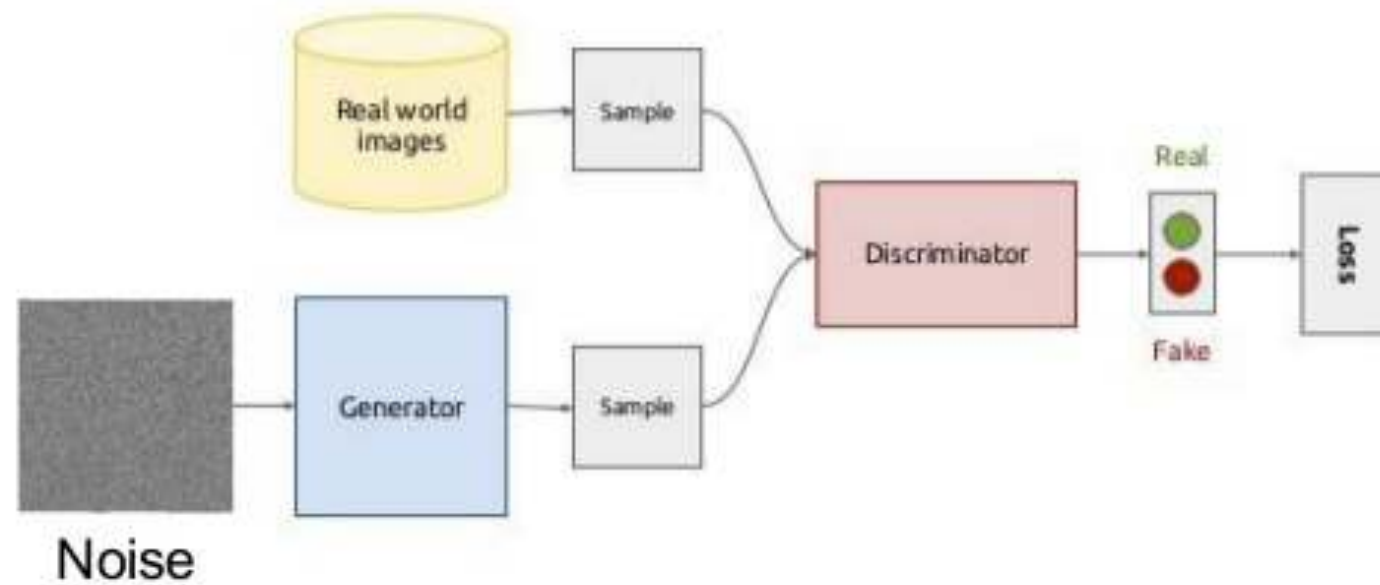
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GANs

Generative Adversarial Nets - Ian et al



5

The GAN algorithm

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations do

for k steps do

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log (1 - D(G(z^{(i)}))) \right].$$

end for

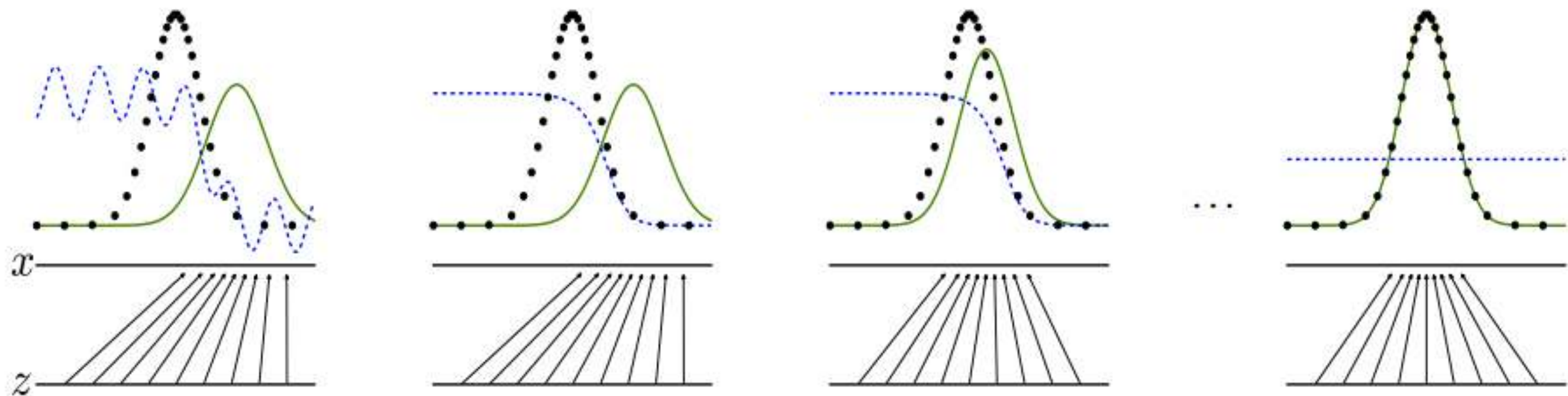
- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log (1 - D(G(z^{(i)}))).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

GANs



GANs

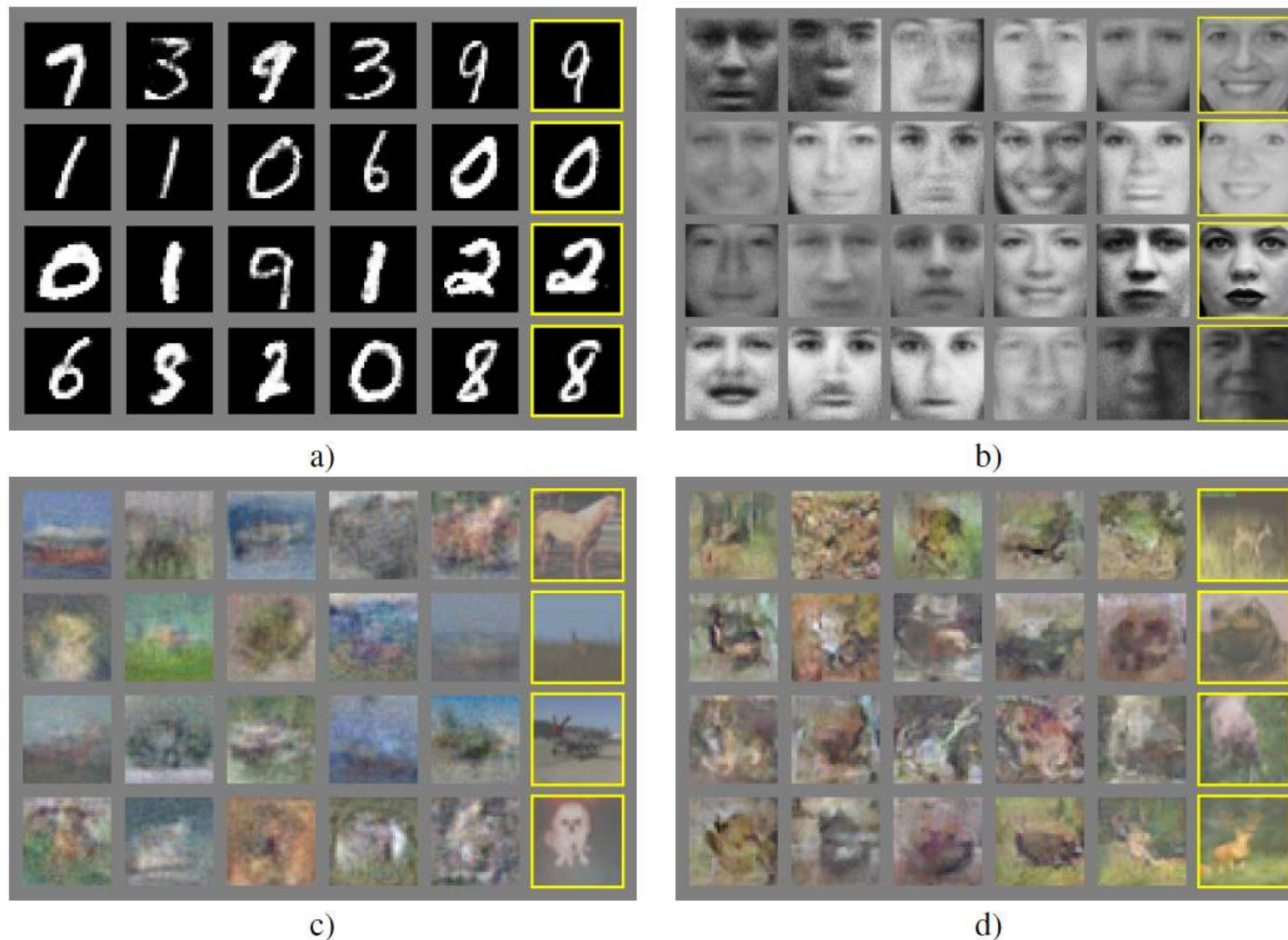
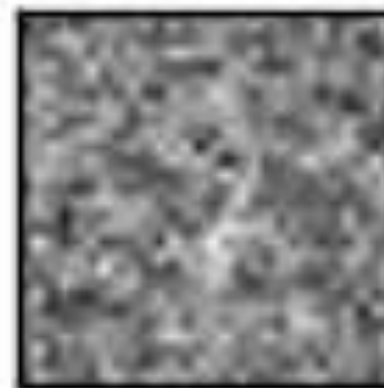
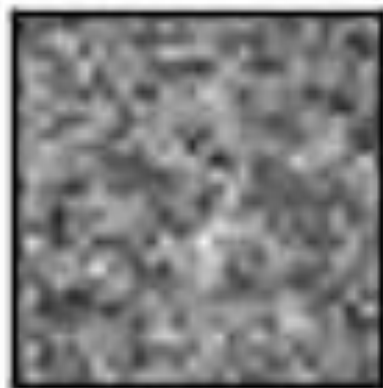
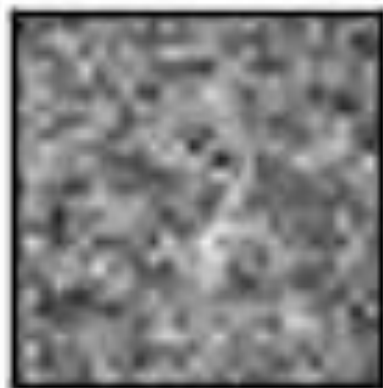


Figure 2: Visualization of samples from the model. Rightmost column shows the nearest training example of the neighboring sample, in order to demonstrate that the model has not memorized the training set. Samples are fair random draws, not cherry-picked. Unlike most other visualizations of deep generative models, these images show actual samples from the model distributions, not conditional means given samples of hidden units. Moreover, these samples are uncorrelated because the sampling process does not depend on Markov chain mixing. a) MNIST b) TFD c) CIFAR-10 (fully connected model) d) CIFAR-10 (convolutional discriminator and “deconvolutional” generator)

GANs



GANs

❖ Pros

- ❖ No inference required or approximations like negative phase of RBMs
- ❖ Model learns the parameters of the distribution and hence does not memorize data.

❖ Cons

- ❖ No explicit expression for the generative distribution.

Deep Learning for Text

Learning Word Representations

Efficient Estimation of Word Representations in Vector Space

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A simple CBOW model

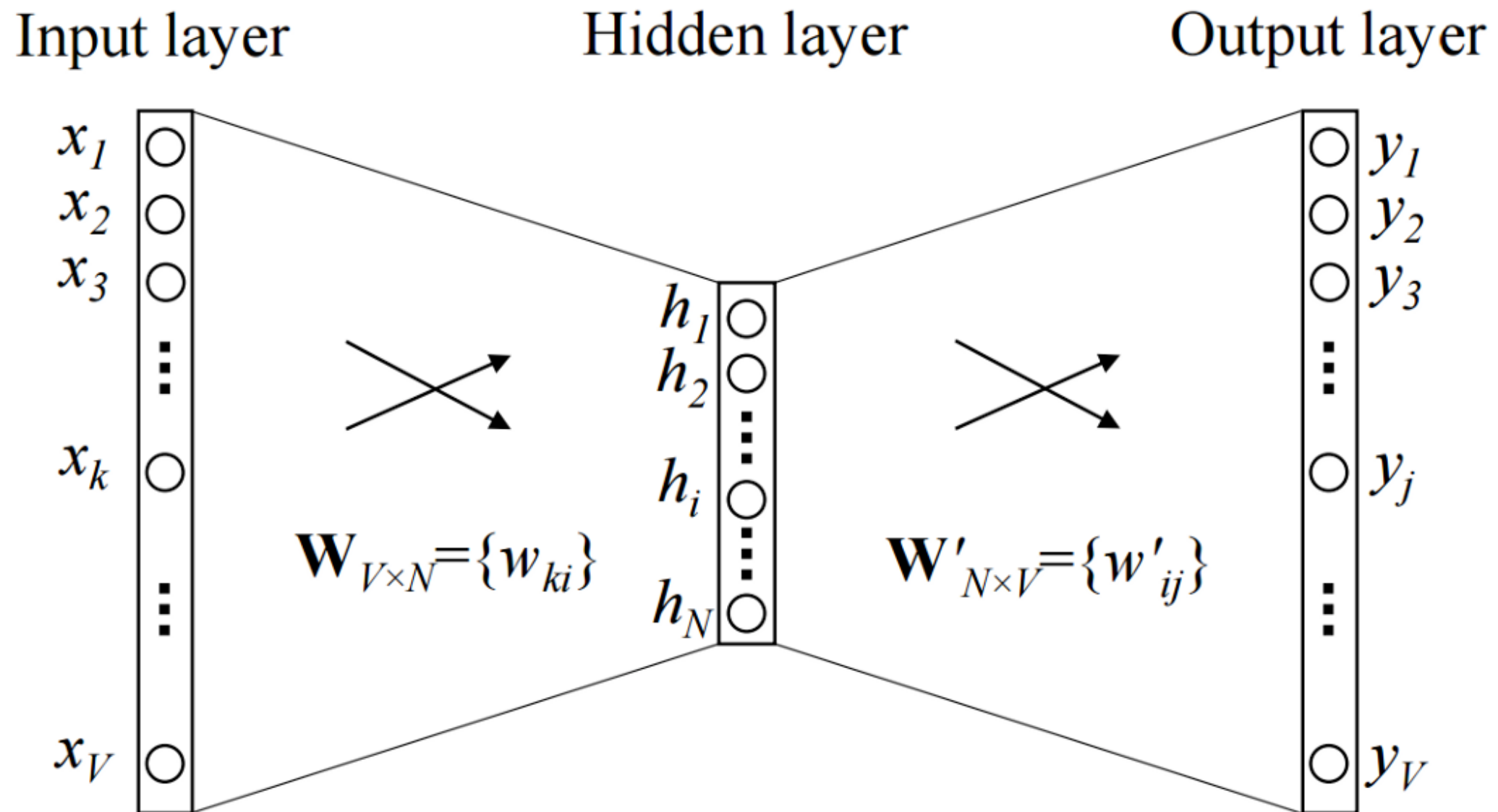
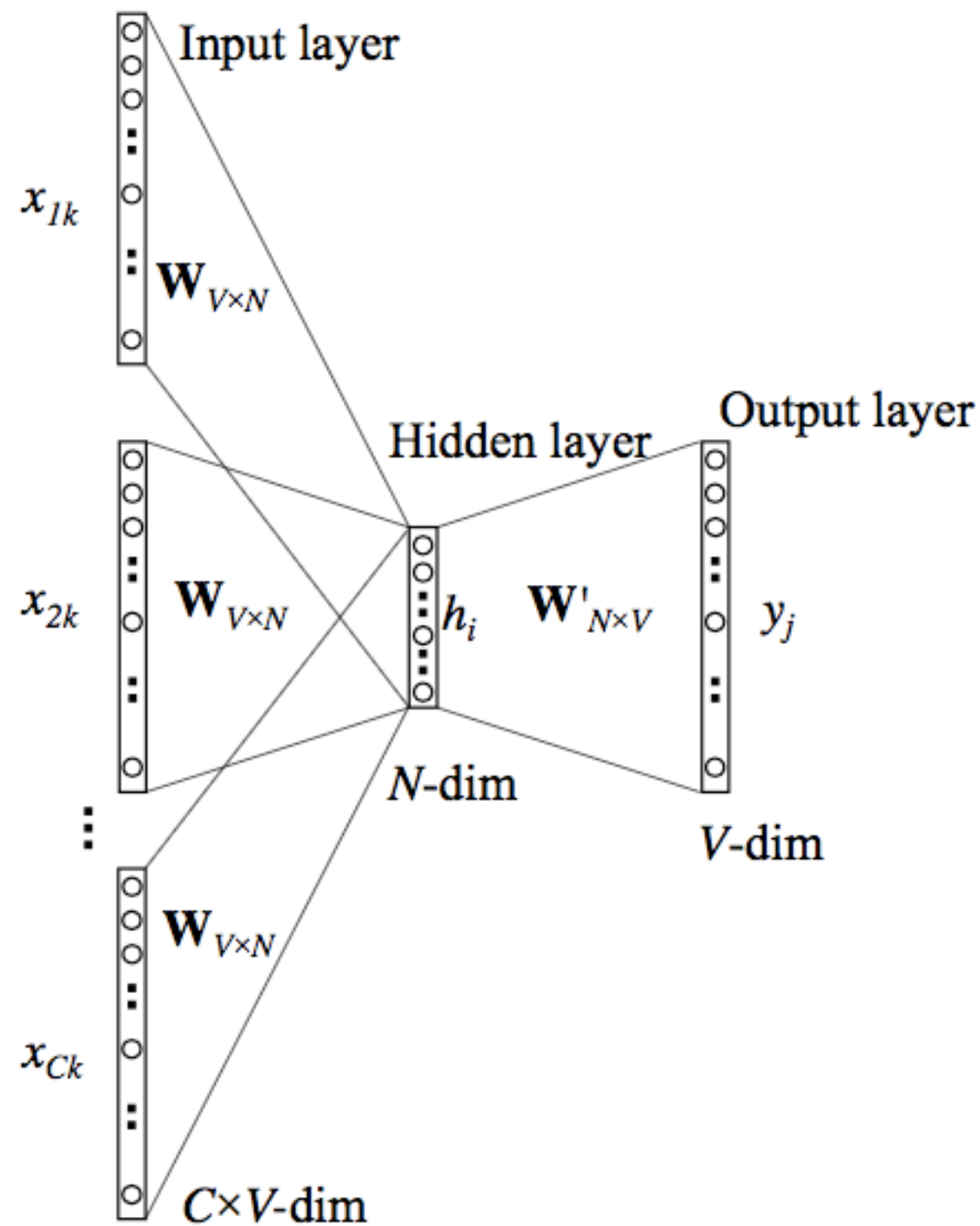
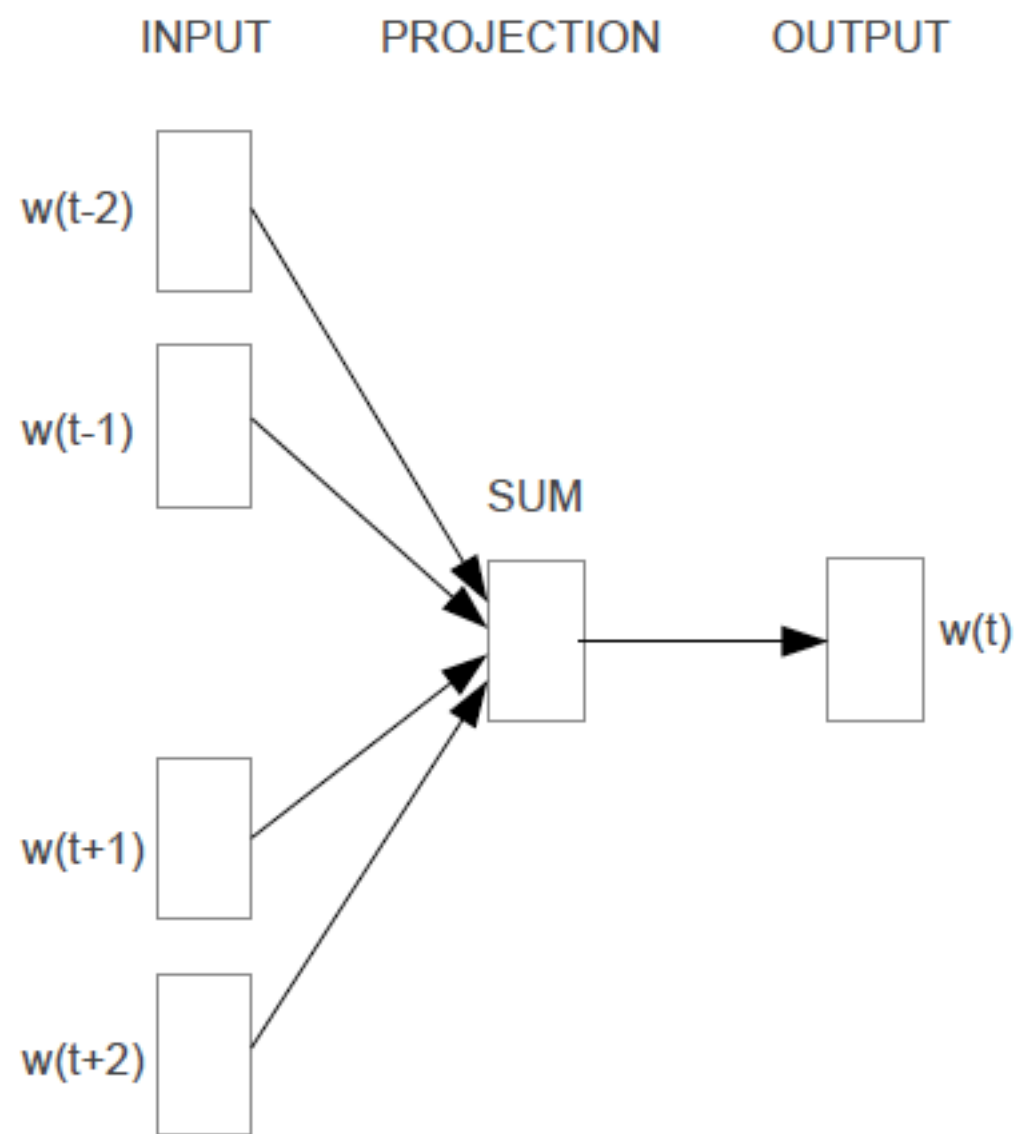


Figure 1: A simple CBOW model with only one word in the context

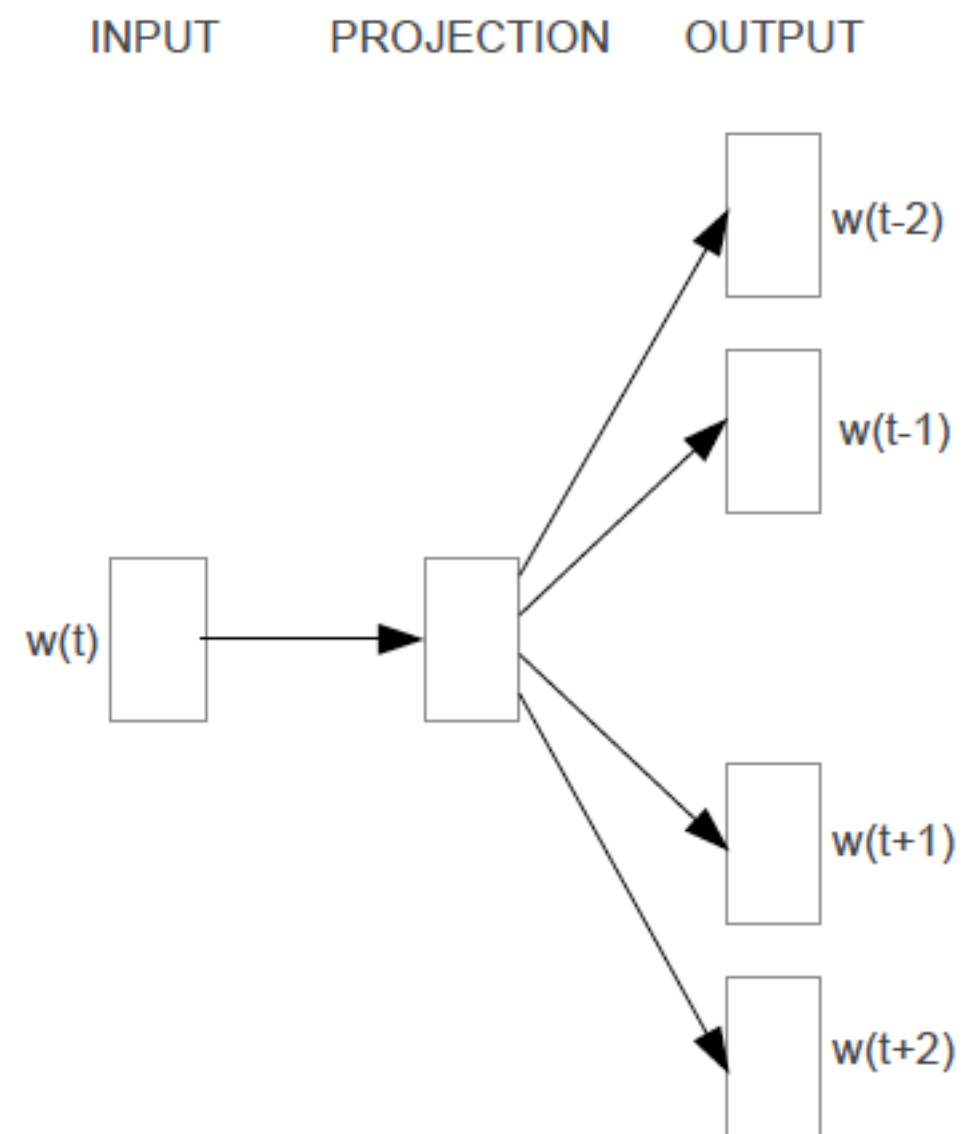
Full CBOW Model



The Two-Models

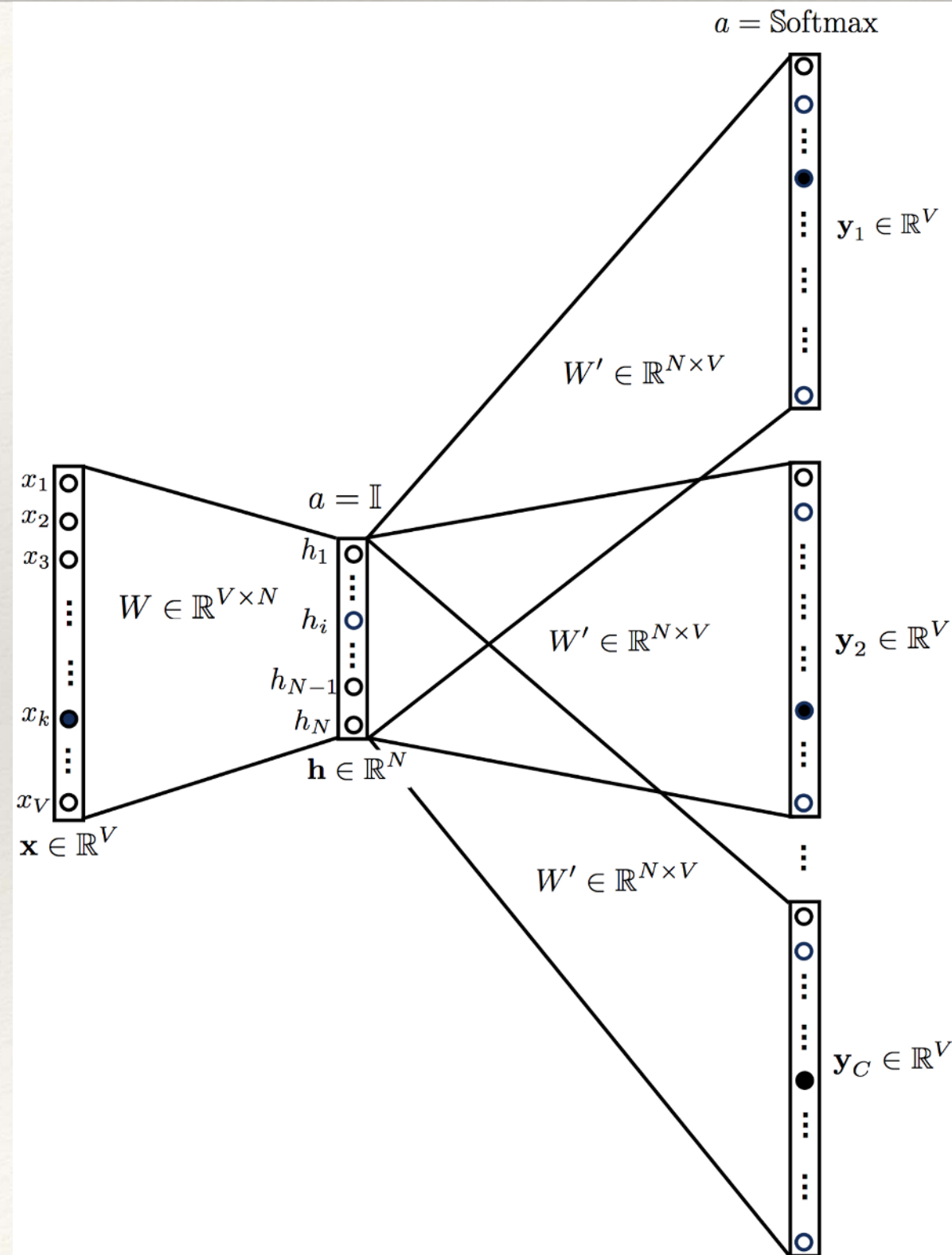


CBOW



Skip-gram

The Skip Gram Model



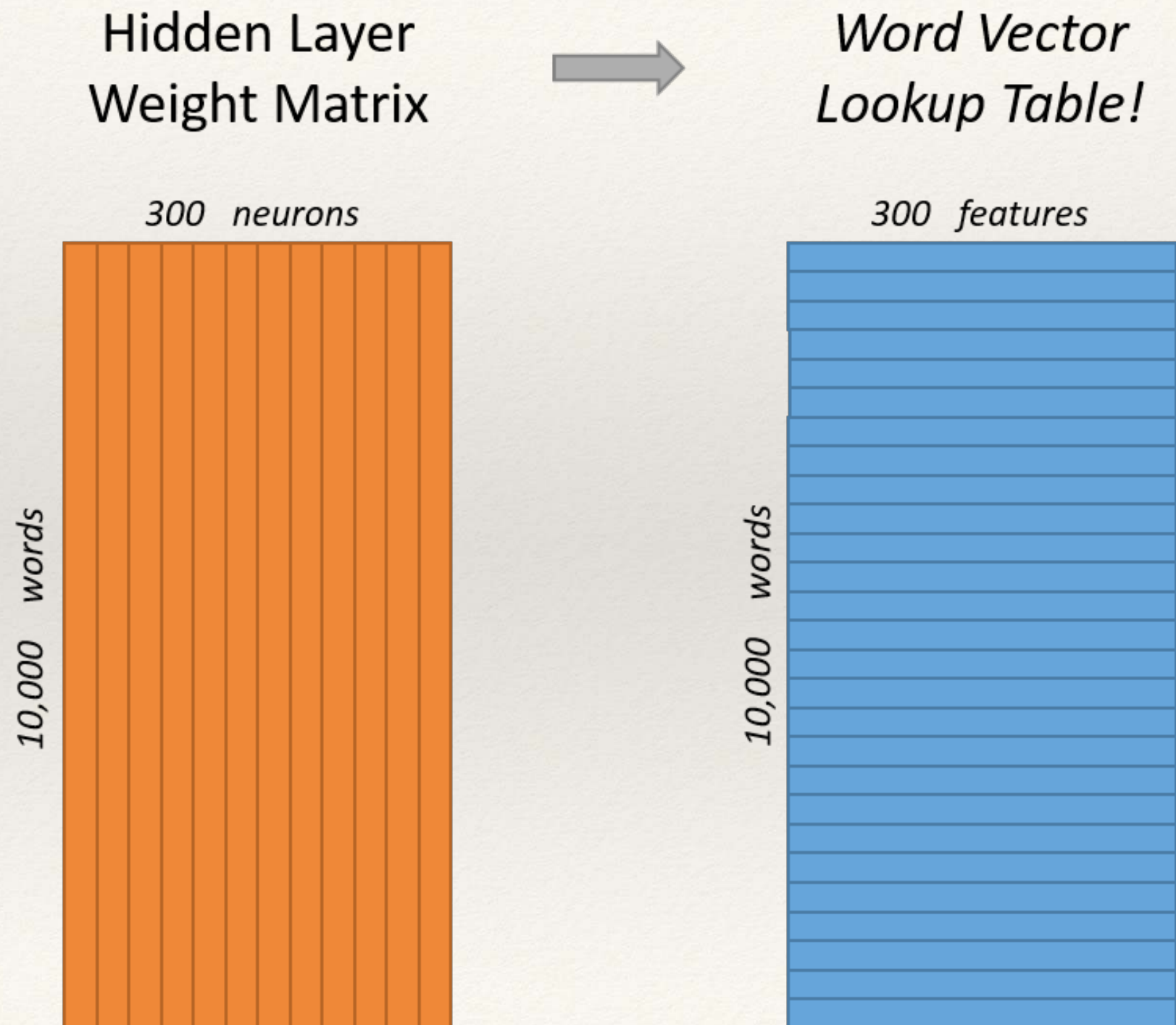
Example given to Skip Gram

Source Text

Training Samples

The quick brown fox jumps over the lazy dog. ➡	(the, quick) (the, brown)
The quick brown fox jumps over the lazy dog. ➡	(quick, the) (quick, brown) (quick, fox)
The quick brown fox jumps over the lazy dog. ➡	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
The quick brown fox jumps over the lazy dog. ➡	(fox, quick) (fox, brown) (fox, jumps) (fox, over)

Skip Gram Model Detailed



Skip Gram Model Detailed

$$[0 \quad 0 \quad 0 \quad \boxed{1} \quad 0] \times \begin{bmatrix} 17 & 24 & 1 \\ 23 & 5 & 7 \\ 4 & 6 & 13 \\ \boxed{10} & \boxed{12} & \boxed{19} \\ 11 & 18 & 25 \end{bmatrix} = [10 \quad 12 \quad 19]$$

First Hidden Layer Output Gives word embeddings after training

Interpreting Word Embeddings

Word	Cosine distance

norway	0.760124
denmark	0.715460
finland	0.620022
switzerland	0.588132
belgium	0.585835
netherlands	0.574631
iceland	0.562368
estonia	0.547621
slovenia	0.531408

Neighbors found for the word “Sweden”

Visualizing Word Embeddings